Diophantine Equations & Integer Restrictions - Teacher Diagnostic Key

Verified answers, source custody and diagnostic routes

RECONNECT -> DISCOVER -> MAKE SENSE -> TRY -> DIAGNOSE -> FADE -> ADOPT -> TRANSFER

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Historical anchor custody

- IOQM-2025-Q03: answer **18**; factor pairs;integer optimization; source status **CLEAN_OFFICIAL**.
- IOQM-2025-Q11: answer **26**; rational approximation;coprime denominator bound; source status **KEY_CORRECTED**. Final official key corrected provisional 61 to 26; independent verification agrees with 26.
- IOQM-2024-Q13: answer **19**; Diophantine factorization;integer system; source status **CLEAN_OFFICIAL**.
- IOQM-2023-Q03: answer **23**; rational approximation;integer inequalities; source status **CLEAN_VALIDATED**.
- IOQM-2023-Q04: answer **07**; Diophantine factorization;divisibility; source status **CLEAN_VALIDATED**. Metadata correction overlay applies: paper has x^4, not x/4; historical source remains clean.
- IOQM-2023-Q11: answer **14**; quadratic Diophantine;factorization;bounds; source status **CLEAN_VALIDATED**.
- IOQM-2023-Q29: answer **95**; sum-product partitions;uniqueness; source status **CLEAN_VALIDATED**.

Practice bank

1. **12**
2. **4**
3. **4**
4. **8**
5. **2**
6. **3**
7. **49**
8. **16**
9. **5**
10. **100**
11. **2**
12. **0**
13. **2**
14. **3**
15. **2**
16. **2**
17. **40**
18. **1**
19. **5**
20. **-28**
21. **7**
22. **4**
23. **4**
24. **21**

Mastery check

1. **29**
2. **8**
3. **31**
4. **2**
5. **7**

6. **12**
7. **58**
8. **12**
9. **14**
10. **14**
11. **2**
12. **96**
13. **11**
14. **67**
15. **2**
16. **25**

Diagnostic routing

- Wrong factor count: check sign/parity/coprime filters and whether cases are exhaustive.
- Correct candidates but wrong final: check reconstruction and substitution into the original condition.
- Long brute force: ask which integer restriction could have been applied before enumeration.
- Rational approximation error: replace decimals by exact cross-products.
- Quadratic integer-feasibility error: distinguish real-root existence from square/discrete feasibility.

All promoted authored numerical answers are recomputed in Authoring/Independent_Math_Audit.md.